

PRIVY: Envisioning and Mitigating Privacy Risks for Consumer-facing AI Product Concepts

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Abstract

AI creates and exacerbates privacy risks, yet practitioners lack effective resources to identify and mitigate these risks. We present PRIVY, a tool that guides practitioners without privacy expertise through structured privacy impact assessments to: (i) identify relevant risks in novel AI product concepts, and (ii) propose appropriate mitigations. PRIVY was shaped by a formative study with 11 practitioners, which informed two versions — one LLM-powered, the other template-based. We evaluated these two versions of PRIVY through a between-subjects, controlled study with 24 separate practitioners, whose assessments were reviewed by 13 independent privacy experts. Results show that PRIVY helps practitioners produce privacy assessments that experts deemed high quality: practitioners identified relevant risks and proposed appropriate mitigation strategies. These effects were augmented in the LLM-powered version. Practitioners themselves rated PRIVY as being useful and usable, and their feedback illustrates how it helps overcome long-standing awareness, motivation, and ability barriers in privacy work.

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CCS Concepts

• **Security and privacy** → **Human and societal aspects of security and privacy**; • **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Privacy, Human-centered AI, Privacy Impact Assessment, Human-AI Collaboration

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1 Introduction

Practitioners building AI products often lack awareness of how privacy risks manifest, are not incentivized to address them, and have few resources to guide effective mitigation [38]. Although prior work has proposed taxonomies to characterize emerging privacy threats in AI systems [40, 61, 72], and broad guidelines for building responsible AI products that address some privacy concerns [7, 8, 47], there remains a critical tooling gap. Practitioners often lack both privacy expertise and the support needed to apply these frameworks to early stage product development. This reflects the broader “principle-practice gap” in ethical and responsible AI — guidelines are plentiful but rarely translate into actionable design processes [58, 73]. In this paper, we ask: **How can we design tools**



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